

Hidden Costs In IoT PRODUCT DEVELOPMENT

Hardware mistakes do not show up early in development. The most expensive hardware mistakes are rarely the obvious ones. Decisions made during design, testing, and architecture shape costs for years.



AKHIL JOY is Director at RIOD and Director of Projects at RNDSquare by RIOD. His areas of focus include Product Engineering, IoT Applications, Vision AI, and EVSE Solutions.

Hardware development is often described as difficult because mistakes made during design can remain hidden until a product reaches manufacturing or even the field. Unlike software, where issues can often be fixed through updates, hardware problems are usually much more expensive to correct once products have been built and deployed. A design decision that appears reasonable during prototyping can later lead to production delays, redesign efforts, supply shortages, or unexpected costs. As a result, understanding the hidden costs of hardware development is essential for design

engineers, product teams, and manufacturers before a product moves beyond the early development stage.

These hidden costs typically emerge across three stages of a product's lifecycle: engineering and design, manufacturing, and field operation. During the engineering phase, teams focus on building prototypes and validating concepts. In manufacturing, the challenge shifts to efficiently scaling production. Once products are deployed, operational issues can introduce additional expenses through maintenance, support, and product updates. Failure to account for risks at any of these stages can lead to lost time, increased costs,